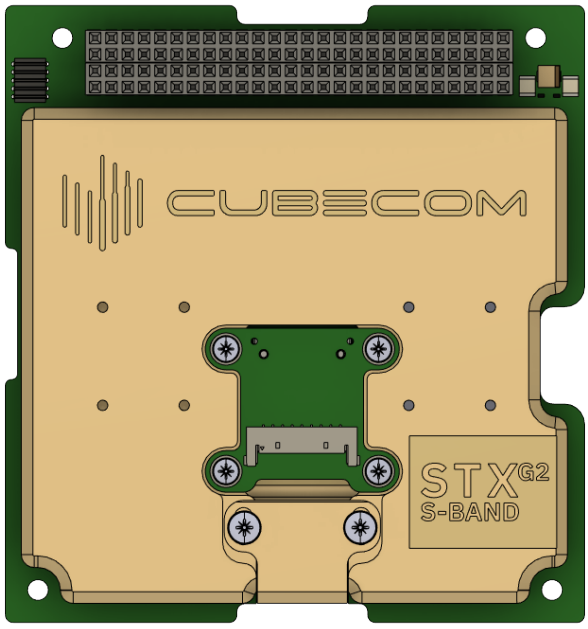
VestaLink S25 (STXG2)

INTERFACE CONTROL DOCUMENT

UID:	ICD-01-10022-A6
Template:	TPL-30-00001-B1
Product UID	01-10022
Product Name	VestaLink S25 (STXG2)
Product Description	Commercial S-Band Nano Satellite Transmitter (25 Mbps; 2.2-2.29 GHz)



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DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Term	Description / Definition
Term	Description / Definition
CAN	Controller Area Network
Complement	Refers to the inverted output of a differential driver
dBm	Decibels relative to one milliwatt
DNC	Do not care
DNP	Do not populate
DVB-S2	Digital Video Broadcasting – Satellite 2
FIFO	First-in, first-out
FPGA	Field-programmable gate array
GHz	Gigahertz
I²C	Inter-Integrated Circuit
LSB	Least significant byte
LVDS	Low Voltage Differential Signaling
Mbps	Megabits per second
MSB	Most significant byte
nCCM	Nano Camera Control Module (from Dragonfly Aerospace cameras)
PPS	Pulse per Second
RF	Radio frequency
SPI	Serial Peripheral Interface
SERDES	Serializer/Deserializer is a highspeed serial communication standard
SpW	SpaceWire
TBC	To be Confirmed
TBD	To be Determined
TM/TC	Telemetry / Telecommand
True	Refers to the non-inverted output of a differential driver
UART	Universal Asynchronous Receiver Transmitter

1 INTRODUCTION AND SCOPE

1.1 IDENTIFICATION

The VestaLink S25 is defined as CUBECOM part number 01-10022. This document is identified as ICD-01-10022.

This product was previously known as the “STXG2” and has been renamed to “VestaLink S25”. The product remains unchanged, with only the product name being updated.

1.2 INTRODUCTION

This document is the Interface Control Document (ICD) for the CUBECOM VestaLink S25 and describes the physical and electrical characteristics of the board interfaces along with the protocols implemented. The VestaLink S25 is the next generation S-Band of the XTX (GEN1).

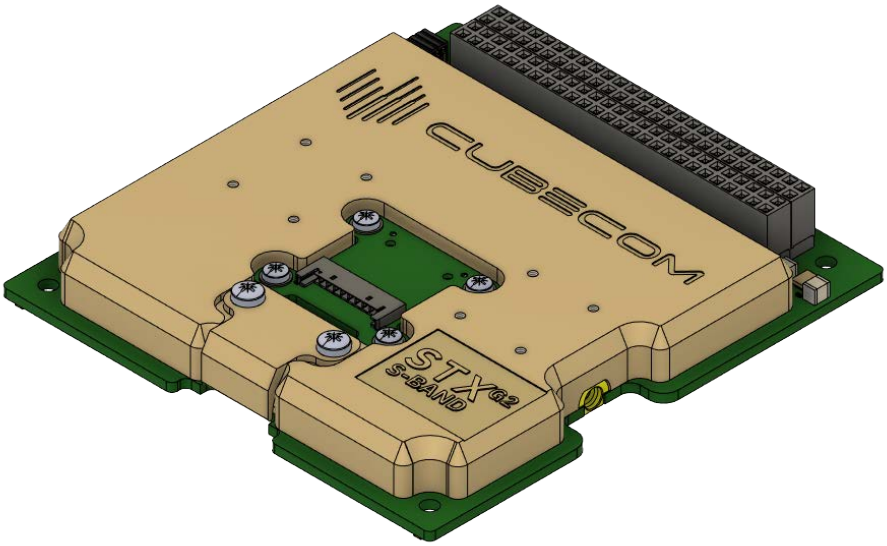


Figure 1-1: VestaLink S25 main board with CSK header.

The VestaLink S25 has multiple interfaces for both TT&C and high-speed data interfaces. The TT&C interfaces are located on the CSK connector while the high-speed interfaces are located on the LSHM connector in the center of the unit.

2 ELECTRICAL INTERFACES

The VestaLink S25 is designed to have a bus interface that is compatible with established nanosat mission integrators. The VestaLink S25 uses up to 12 W depending on the RF power level selected.

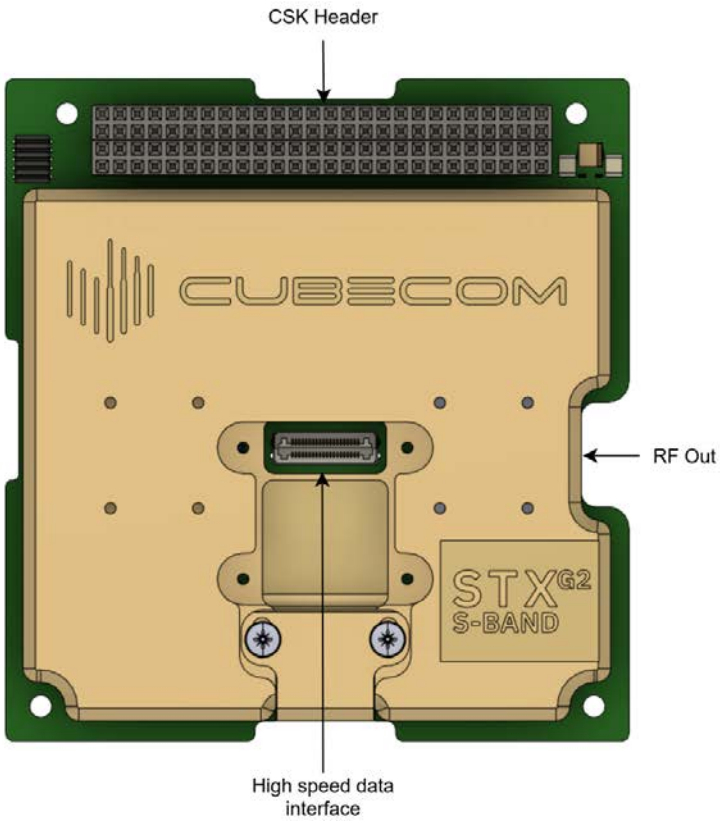


Figure 2-1: Electrical interface overview - PC104 connector option.

This chapter describes the power, TM/TC and data interfaces available on the VestaLink S25.

2.1 CSK HEADER (TM/TC AND POWER INTERFACES)

The CSK header implements the standard PC104 layout used in the original Pumpkin CubeSat standard and provides the power and control interfaces for the transmitter. The fitted header's part number is Samtec ESQ-126-39-G-D. The pin selection on the CSK header is done on the option sheet when ordering the product.

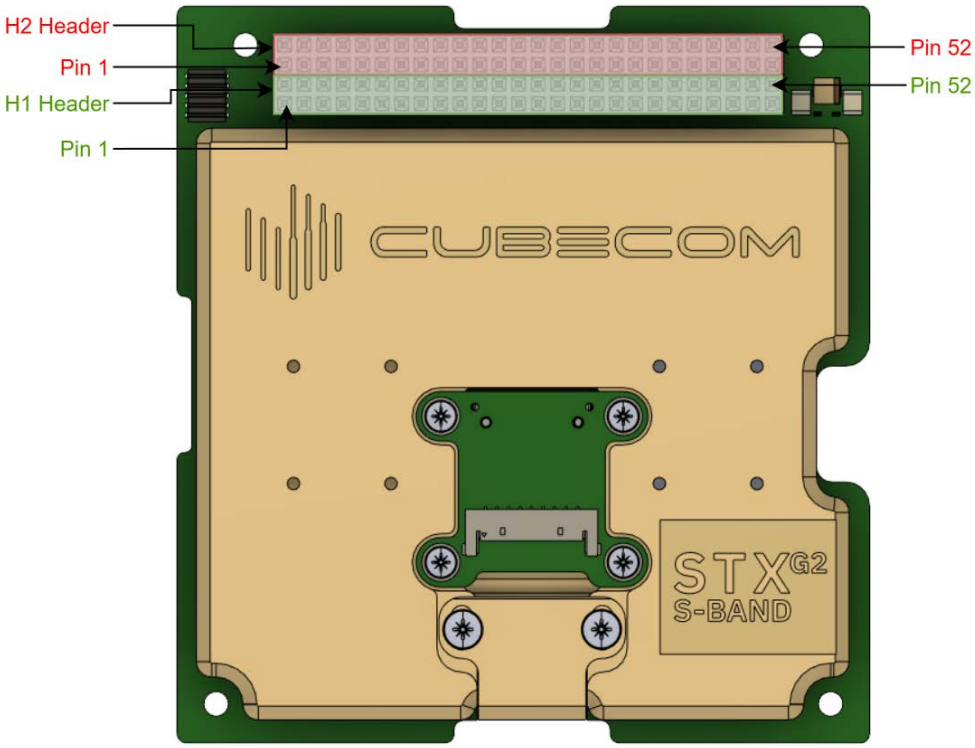


Figure 2-2: PC104-header pin numbering.

Table 2-1: H1 and H2 pin options.

Signal Name	Primary Pin	Alternate	I/O Type	Description
VBAT	H2.45, H2.46	H2.11	Power	Unregulated battery bus voltage input 7V-18V
GND	H2.29, H2.30, H2.32	H2.14	Power	Power ground
SDA0	H1.41	H1.45	Bidirectional	I^2C serial data
SCL0	H1.43	H1.47	Bidirectional	I^2C serial clock
CANH	H1.42,	H1.3	Bidirectional	CAN Hi
CANL	H1.44	H1.1	Bidirectional	CAN Lo
UART TX	H1.33	H1.35, H1,39, H1,40, H2.21, H2.22	Output	UART transmit
UART RX	H1.35	H1.33, H1,39, H1.40, H2.21, H2.22	Input	UART receive
SPI_SCLK	H1.29	H1.21	Input	SPI clock pin (data interface)
SPI_MOSI	H1.30	H1.23	Input	SPI data pin (data interface)
SPI_CS	H1.9	H1.10, H1.11, H1.13, H1.14, H1.32	Input	SPI chip-select pin (data interface)
SPI_TR	H1.7	H2.50	Output	(Transmit ready) SPI flow control pin
nFPGA_RESET	H1.8	H1.49	Input	FPGA reset (active low, optional)
ENABLE	H1.18	H1.4	Input	Board Enable (optional)

Notes:

- Pins H2.45 and H2.46 are tied together even if alternate H2.11 pin was selected on the option sheet.
- Pins H2.29, H2.30 and H2.32 are tied together even if alternate H2.14 pin was selected on the option sheet.

2.2 RADIO CONTROL

The VestaLink S25 has two control signals, namely Enable and nReset, if chosen in the option sheet, that are used to enable/reset the radio. The VestaLink S25 Enable signal enables the power system of the board. The nReset signal controls the reset state of the FPGA/Processor. LOW resets the board, and HIGH keeps the board in its active state. Resetting the system will set internal logic/registers to their default values.

Table 2-2: Radio control electrical specifications

Symbol	Description	Min	Typ	Max	Units
V_{EN_IH}	Input voltage level to enable VestaLink S25	1.25	3.3	3.6	V
V_{EN_IL}	Input voltage level to disable VestaLink S25	-	0	0.75	V
I_{EN}	Current draw at 3.3V when enabled		250		uA
V_{NRST_IH}	Input high voltage level for NRST (active low).	2	3.3	3.6	V
V_{NRST_IL}	Input low voltage level for NRST (active low).	-	0	0.8	V

2.3 I²C INTERFACE

The I²C interface offers a standardized I²C interface. An I²C isolator (buffer) is present on the bus side. The I²C electrical characteristics can be found in Table 2-3.

Table 2-3: I²C electrical specifications.

Symbol	Description	Min	Typ	Max	Units
V_{MAX}	Maximum input voltage on SDA and SCL line	-	-	5.5	V
V_{PULL}	Pull-up voltage	-	3.3	-	V
I_{LI}	SDA, SCL leakage current (SDA, SCL = 5.5V)	-1	-	1	µA
V_{OL}	Low-level output voltage (I_{OL} = 6mA)	0.47	0.52	0.6	V
F_{SCL}	Clock rate of SCL line	-	-	400	kHz

2.4 CAN INTERFACE

The VestaLink S25 is equipped with a CAN interface to facilitate TM/TC (Telemetry/Telecommand) operations. These interfaces can be configured to operate in redundant mode such that if the primary interface fails, the secondary interface can be used without significant system impact.

Table 2-4: CAN bus electrical specifications.

Symbol	Description	Min	Typ	Max	Units
V_{ODIFF}	Differential output voltage ($Q - \bar{Q}$), R_{DIFF} = 50Ω (dominant mode)	1.5	2.2	3.0	V
V_{OCM}	Output common-mode voltage (dominant mode)	1.45	1.95	2.45	V
V_{ITH+}	Receiver Differential Input Threshold Voltage (going positive)	-	0.775	0.9	V
V_{ITH-}	Receiver Differential Input Threshold Voltage (going negative)	0.5	0.625		V
V_{IHYST}	Receiver Input Hysteresis	-	25	-	mV
V_{IMAX}	Absolute maximum voltage limits on bus	-25	-	25	V

2.5 SPI INTERFACE

The SPI interface offers a standardized SPI interface, along with an integrated bus buffer. The SPI electrical characteristics can be found in Table 2-5.

Table 2-5: SPI electrical specifications.

Symbol	Description	Min	Typ	Max	Units
V_{MAX}	Maximum input voltage on TX and RX line	-	-	5.5	V
V_O	Output voltage (V_{CC} = 3.3V)	-	3.3	-	V
V_{IL}	Low-level input voltage (V_{CC} = 3.3V)	-	0.8	-	V
V_{IH}	High-level input voltage (V_{CC} = 3.3V)	-	2.0	-	V

2.6 UART INTERFACE

The UART interface offers a standardized UART interface, along with an integrated bus buffer. The UART electrical characteristics can be found in Table 2-6.

Table 2-6: UART electrical specifications.

Symbol	Description	Min	Typ	Max	Units
V_{MAX}	Maximum input voltage on TX and RX line	-	-	5.5	V
V_O	Output voltage (V_{CC} = 3.3V)	-	3.3	-	V
V_{IL}	Low-level input voltage (V_{CC} = 3.3V)	-	0.8	-	V
V_{IH}	High-level input voltage (V_{CC} = 3.3V)	-	2.0	-	V

2.7 RS485/RS422

The RS485 and RS422 interfaces have been defined, and the hardware already supports these TT&C interfaces. The protocol is still TBC. Contact CUBECOM if this interface is required. The pins are shared which means either the RS485 or RS422 interface can be utilized at a time.

Table 2-7: RS485 & RS422 electrical specifications.

Symbol	Description	Min	Typ	Max	Units
V_{ODIFF}	Differential output voltage ($Q - \bar{Q}$), R_{DIFF} = 100Ω	2	2.4	3.3	V
V_{OCM}	Output common-mode voltage	-	-	3	V
V_{ITH}	Receiver Differential Input Threshold Voltage	-	-	±0.2	V
V_{IHYST}	Receiver Input Hysteresis	-	25	-	mV
V_{IMAX}	Absolute maximum voltage limits on bus	-11	-	15	V

2.8 HIGH-SPEED DATA INTERFACE

There are eight LVDS pairs available on the high-speed LSHM connector to implement the following interfaces:

- SpaceWire
- CUBECOM LVDS
- SPI
- Custom data interface (on request)

There are also two XCVR receiver pairs available to implement the SERDES interface on the LSHM connector.

Note that only one high-speed protocol can be chosen at a time on the VestaLink S25. This will be chosen using the Option Sheet.

A slower (10 MHz) data interface is also available in the form of an SPI interface, which is implemented on the CSK connector.

The connector facilitating the LVDS pairs on the board is a Samtec LSHM-120-02.5-L-DV-A-S-K-TR.

The differential impedance of the LVDS lines is 100 Ω.

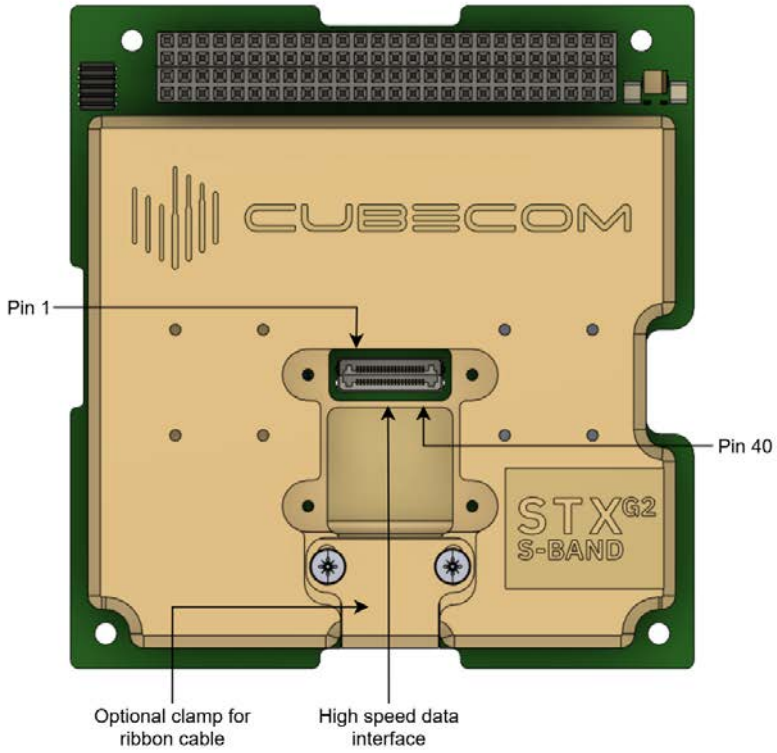


Figure 2-3: VestaLink S25 high-speed data interface.

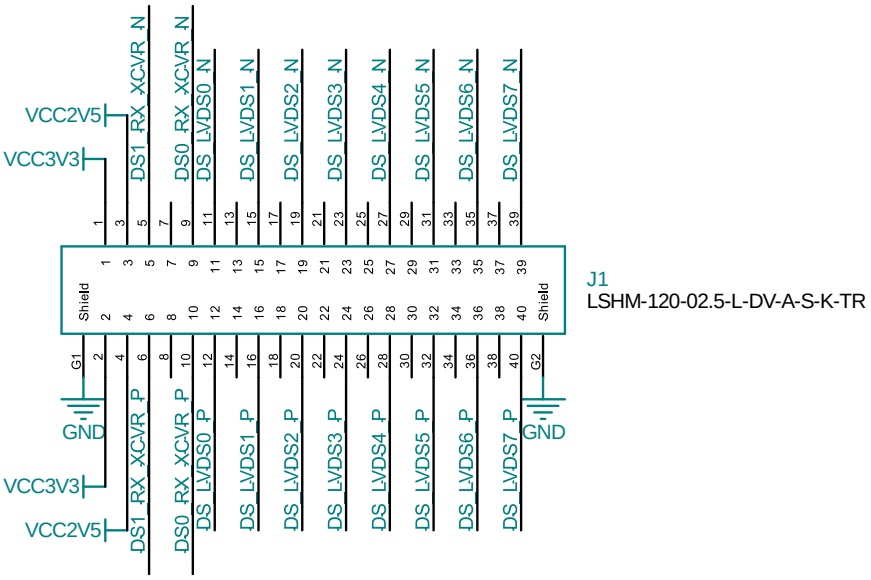


Figure 2-4:LHSM connector configuration for custom interface applications.

Table 2-8:LHSM connector description for custom interface applications.

Pin Nr	Signal	Description
1	VCC3V3	3.3V supply capable of powering external logic buffers
2	VCC3V3	3.3V supply capable of powering external logic buffers
3	VCC2V5	2.5V supply capable of powering external logic buffers
4	VCC2V5	2.5V supply capable of powering external logic buffers
5	DS1_RX_XCVR_N	Gigabit capable receive pin for high-speed interfaces (Negative polarity)
6	DS1_RX_XCVR_P	Gigabit capable receive pin for high-speed interfaces (Positive polarity)
7	Reserved	Gigabit capable transmit pin allocated to this location
8	Reserved	Gigabit capable transmit pin allocated to this location
9	DS0_RX_XCVR_N	Gigabit capable receive pin for high-speed interfaces (Negative polarity)
10	DS0_RX_XCVR_P	Gigabit capable receive pin for high-speed interfaces (Positive polarity)
11	DS_LVDS0_N	LVDS0 Negative Input or Output
12	DS_LVDS0_P	LVDS0 Positive Input or Output
13	Open	
14	Open	
15	DS_LVDS1_N	LVDS1 Negative Input or Output
16	DS_LVDS1_P	LVDS1 Positive Input or Output
17	Open	
18	Open	
19	DS_LVDS2_N	LVDS2 Negative Input or Output

Pin Nr	Signal	Description
20	DS_LVDS2_P	LVDS2 Positive Input or Output
21	Open	
22	Open	
23	DS_LVDS3_N	LVDS3 Negative Input or Output
24	DS_LVDS3_P	LVDS3 Positive Input or Output
25	Open	
26	Open	
27	DS_LVDS4_N	LVDS4 Negative Input or Output
28	DS_LVDS4_P	LVDS4 Positive Input or Output
29	Open	
30	Open	
31	DS_LVDS5_N	LVDS5 Negative Input or Output
32	DS_LVDS5_P	LVDS5 Positive Input or Output
33	Open	
34	Open	
35	DS_LVDS6_N	LVDS6 Negative Input or Output
36	DS_LVDS6_P	LVDS6 Positive Input or Output
37	Open	
38	Open	
39	DS_LVDS7_N	LVDS7 Negative Input or Output
40	DS_LVDS7_P	LVDS7 Positive Input or Output

Table 2-9: LVDS electrical specifications

Symbol	Description	Min	Typ	Max	Units
V_{ODIFF}	Differential output voltage $(Q - \bar{Q})$	250	350	450	mV
V_{OCM}	Output common-mode voltage	1.125	1.2	1.375	V
V_{IDIFF}	Differential input voltage $(Q - \bar{Q})$	100	350	600	mV
V_{ICM_DC}	Input common-mode voltage (DC coupling)	0.6	1.25	2.35	V

It should be noted that there are no ground pins available on the LSHM connector. The ground is transferred via the shielding tabs on the connector.

2.8.1 SPACEWIRE

The SpaceWire interfaces (SpaceWire 0 and SpaceWire 1) comply with ECSS-E-ST-50-12C, apart from the physical connector. The interface has the capability to operate at a maximum speed of 200 Mbps (Raw data throughput of 160 Mbps). After enabling the node, the interface is set to receive NULL characters, at a rate of 10 Mbps, to establish a connection. Once a successful connection is established, the transmit bit rate can be adjusted to a maximum of 200 Mbps. The receive bit rate can be configured to any value within the range of 0-200 Mbps.

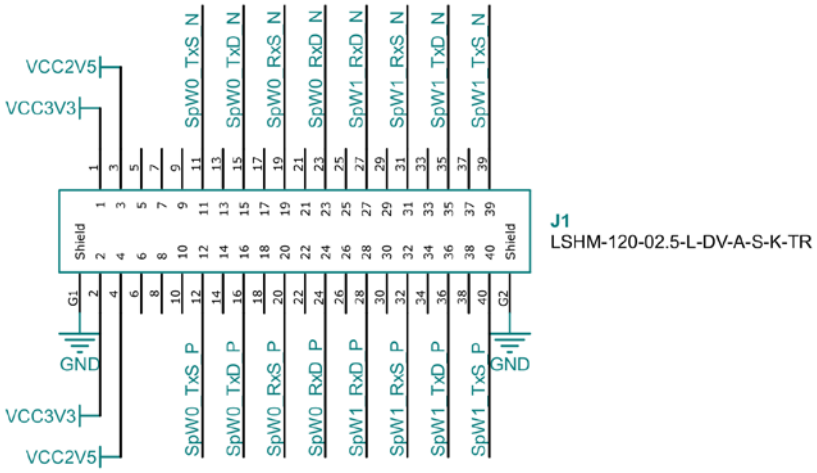


Figure 2-5: SpaceWire pinout on LSHM connector.

Table 2-10: SpaceWire pinouts used on LSHM connector.

Pin Nr	Signal	Description
11	SpW0_TxS_N	SpaceWire 0 Strobe Transmit Compliment (-) (DS_LVDS0_N)
12	SpW0_TxS_P	SpaceWire 0 Strobe Transmit True (+) (DS_LVDS0_P)
15	SpW0_TxD_N	SpaceWire 0 Data Transmit Compliment (-) (DS_LVDS1_N)
16	SpW0_TxD_P	SpaceWire 0 Data Transmit True (+) (DS_LVDS1_P)
19	SpW0_RxS_N	SpaceWire 0 Strobe Receive Compliment (-) (DS_LVDS2_N)
20	SpW0_RxS_P	SpaceWire 0 Strobe Receive True (+) (DS_LVDS2_P)
23	SpW0_RxD_N	SpaceWire 0 Data Receive Compliment (-) (DS_LVDS3_N)
24	SpW0_RxD_P	SpaceWire 0 Data Receive True (+) (DS_LVDS3_P)
27	SpW1_RxD_N	SpaceWire 1 Data Receive Compliment (-) (DS_LVDS4_N)
28	SpW1_RxD_P	SpaceWire 1 Data Receive True (+) (DS_LVDS4_P)
31	SpW1_RxS_N	SpaceWire 1 Strobe Receive Compliment (-) (DS_LVDS5_N)
32	SpW1_RxS_P	SpaceWire 1 Strobe Receive True (+) (DS_LVDS5_P)
35	SpW1_TxD_N	SpaceWire 1 Data Transmit Compliment (-) (DS_LVDS6_N)
36	SpW1_TxD_P	SpaceWire 1 Data Transmit True (+) (DS_LVDS6_P)
39	SpW1_TxS_N	SpaceWire 1 Strobe Transmit Compliment (-) (DS_LVDS7_N)

Pin Nr	Signal	Description
40	SpW1_TxS_P	SpaceWire 1 Strobe Transmit True (+) (DS_LVDS7_P)
Shield	Ground	Ground reference is available from the connector shield

A daughterboard is shipped with the VestaLink S25 to easily connect the SpaceWire interface to a harness. Connector J2 and J3 on the daughterboard are Molex Pico Lock 5040500991 connectors. Connector J2 (on top of the daughterboard) is connected to SpaceWire interface 0, and J3 (on bottom of the daughterboard) is connected to SpaceWire interface 1. SpaceWire interface 0 is the primary interface with interface 1 being the backup interface.

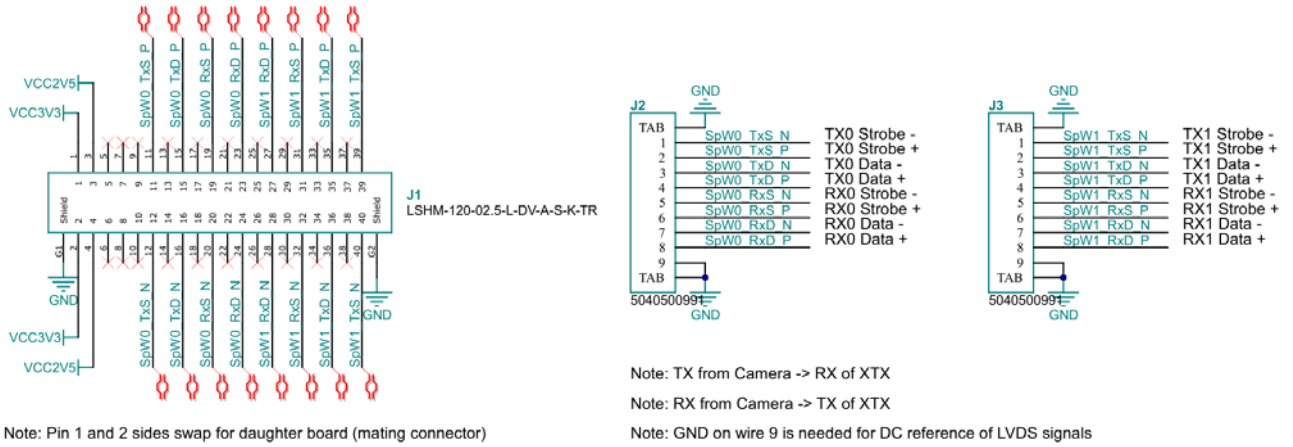


Figure 2-6: SpaceWire interface using LSHM daughter board schematic.

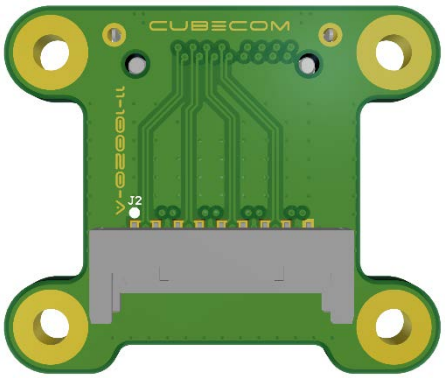


Figure 2-7: LSHM daughter board Top View

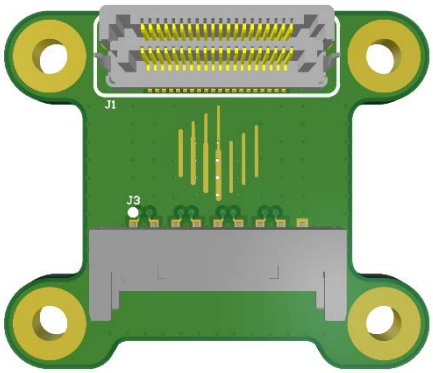


Figure 2-8: LSHM daughter board Bottom View.

The PCB information for this daughter board is available to customers to implement their own variations with custom connectors or to add logic translators for specific IO requirements. 3.3V and 2.5V power is available to power the logic translators.

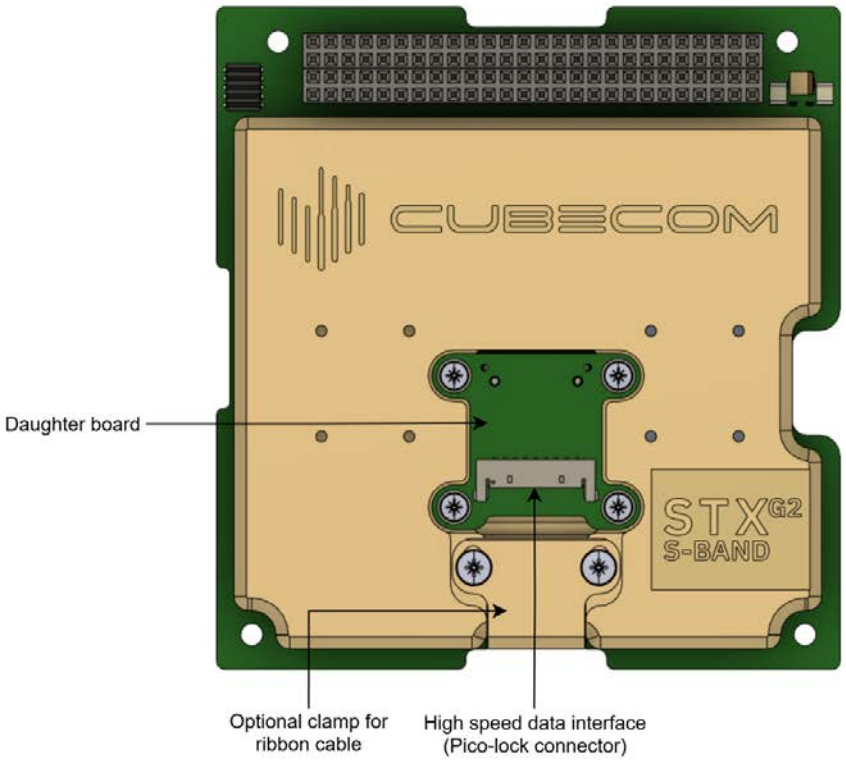


Figure 2-9: LSHM daughter board fitted to VestaLink S25.

2.8.2 CUBECOM LVDS INTERFACE

The CUBECOM LVDS interfaces (LVDS 0 and LVDS 1) are available to directly interface the VestaLink S25 to a data source configured as a CUBECOM LVDS transmitter node. The LVDS receiver node runs at 200 MHz allowing for a LVDS transmitter node to run at any speed up to and including 200 MHz. The system uses a 10-bit packet containing 8-bits of data. The raw data throughput is therefore 4/5 of the data rate.

Table 2-11: Effective LVDS data rates.

Channel	Mode	Clock Speed (MHz)	Line Rate (Mbps)	Data Rate (Mbps)
D0	SDR	150	150	120
	DDR	150	300	240
D1	SDR	150	150	120
	DDR	150	300	240
Both	SDR	150	300	240
	DDR	150[1]	600	480

The pinout of the high-speed connector used for the CUBECOM LVDS interface is shown in Figure 2-10.

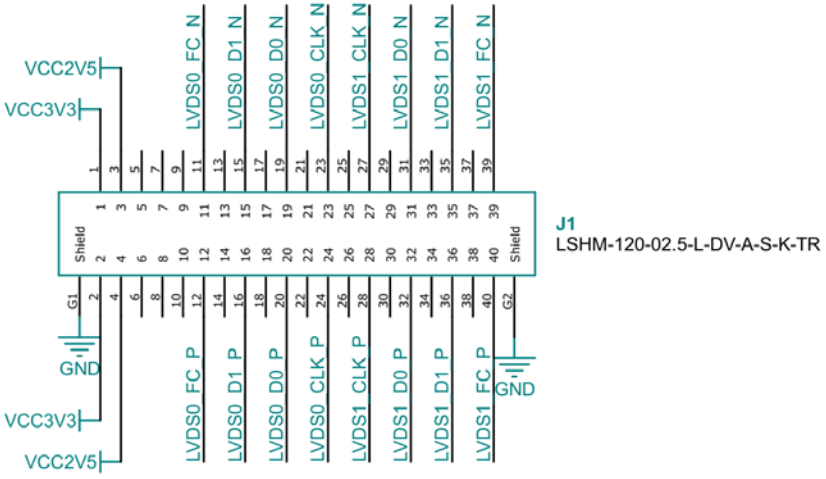


Figure 2-10: LVDS pinout on LHSM connector.

[1] Highest clock speed supported for dual-node, dual data-rate is 150 MHz.

Table 2-12: LVDS connector description for CUBECOM LVDS interface.

Pin Nr	Signal	Description
11	LVDS0_FC_N	LVDS 0 Output Flow Control Compliment (-) (DS_LVDS0_N)
12	LVDS0_FC _P	LVDS 0 Output Flow Control True (+) (DS_LVDS0_P)
15	LVDS0_D1_N	LVDS 0 Input Data Lane 1 Compliment (-) (DS_LVDS1_N)
16	LVDS0_D1_P	LVDS 0 Input Data Lane 1 True (+) (DS_LVDS1_P)
19	LVDS0_D0_N	LVDS 0 Input Data Lane 0 Compliment (-) (DS_LVDS2_N)
20	LVDS0_D0_P	LVDS 0 Input Data Lane 0 True (+) (DS_LVDS2_P)
23	LVDS0_CLK_N	LVDS 0 Input LVDS Clock Compliment (-) (DS_LVDS3_N)
24	LVDS0_CLK_P	LVDS 0 Input LVDS Clock True (+) (DS_LVDS3_P)
27	LVDS1_CLK_N	LVDS 1 Input LVDS Clock Compliment (-) (DS_LVDS4_N)
28	LVDS1_CLK_P	LVDS 1 Input LVDS Clock True (+) (DS_LVDS4_P)
31	LVDS1_D0_N	LVDS 1 Input Data Lane 0 Compliment (-) (DS_LVDS5_N)
32	LVDS1_D0_P	LVDS 1 Input Data Lane 0 True (+) (DS_LVDS5_P)
35	LVDS1_D1_N	LVDS 1 Input Data Lane 1 Compliment (-) (DS_LVDS6_N)
36	LVDS1_D1_P	LVDS 1 Input Data Lane 1 True (+) (DS_LVDS6_P)
39	LVDS1_FC_N	LVDS 1 Output Flow Control Compliment (-) (DS_LVDS7_N)
40	LVDS1_FC_P	LVDS 1 Output Flow Control True (+) (DS_LVDS7_P)
Shield	Ground	Ground reference is available from the connector shield

The same daughterboard used for SpaceWire (shown in Figure 2-7) can also be used for the CUBECOM LVDS interface. The interface pinout of the daughterboard is shown in Figure 2-11.

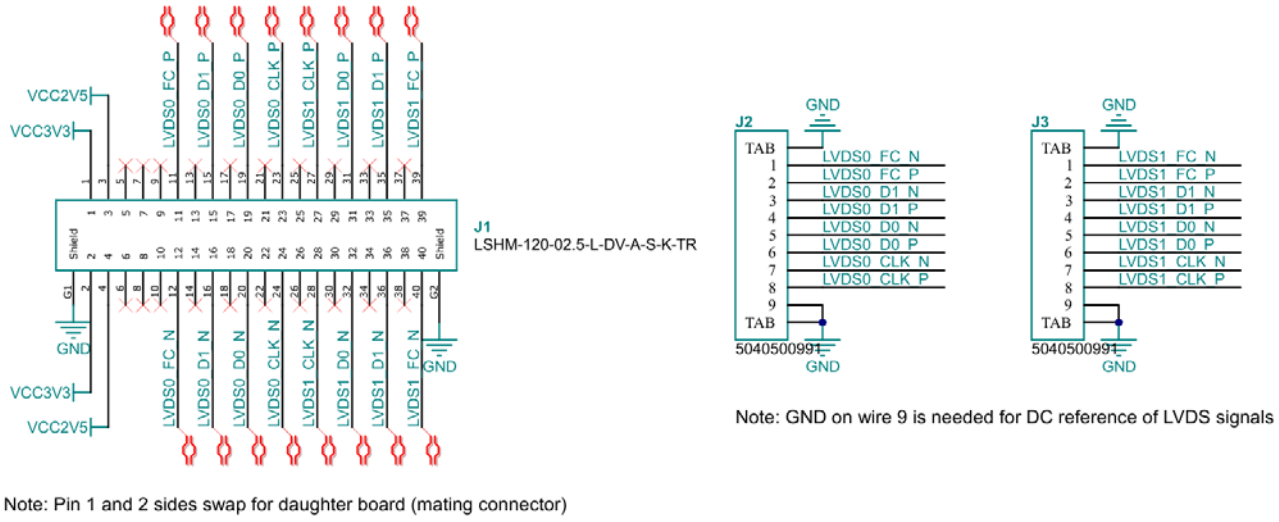


Figure 2-11: LVDS interface using LSHM daughter board schematic.

2.8.3 SPI INTERFACE

The SPI interfaces (SPI 0 and SPI 1) are available to directly interface the VestaLink S25 to a data source using a SPI output. The maximum rate the SPI can operate at is 10 MHz

The pinout of the high-speed connector used for the SPI interface is shown in Figure 2-10.

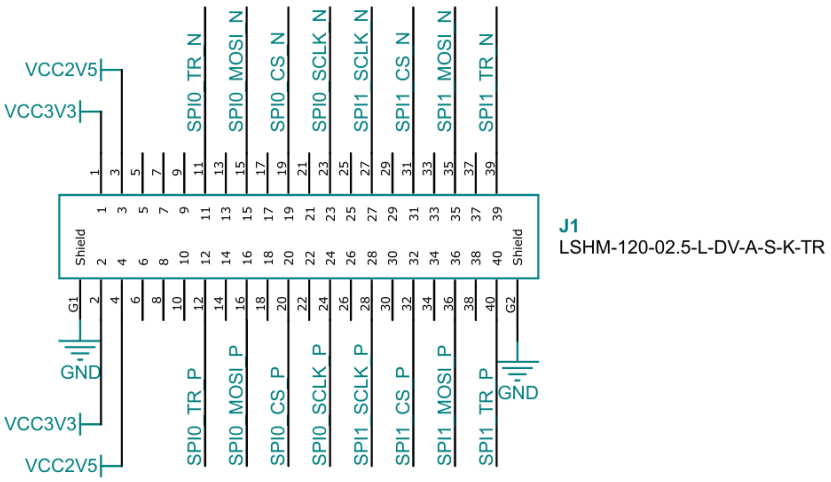


Figure 2-12: SPI pinout on LHSM Connector.

Table 2-13: LVDS connector description for SPI interface.

Pin Nr	Signal	Description
11	SPI0_TR_N	SPI 0 Output Transmit Ready Compliment (-) (DS_LVDS0_N)
12	SPI0_TR_P	SPI 0 Output Transmit Ready True (+) (DS_LVDS0_P)
15	SPI0_MOSI_N	SPI 0 Input MOSI Compliment (-) (DS_LVDS1_N)
16	SPI0_MOSI_P	SPI 0 Input MOSI 1 True (+) (DS_LVDS1_P)
19	SPI0_CS_N	SPI 0 Input Chip Select 0 Compliment (-) (DS_LVDS2_N)
20	SPI0_CS_P	SPI 0 Input Chip Select 0 True (+) (DS_LVDS2_P)
23	SPI0_SCLK_N	SPI 0 Input SPI Clock Compliment (-) (DS_LVDS3_N)
24	SPI0_SCLK_P	SPI 0 Input SPI Clock True (+) (DS_LVDS3_P)
27	SPI1_SCLK_N	SPI 1 Input SPI Clock Compliment (-) (DS_LVDS4_N)
28	SPI1_SCLK_P	SPI 1 Input SPI Clock True (+) (DS_LVDS4_P)
31	SPI1_CS_N	SPI 1 Input Chip Select 0 Compliment (-) (DS_LVDS5_N)
32	SPI1_CS_P	SPI 1 Input Chip Select 0 True (+) (DS_LVDS5_P)
35	SPI1_MOSI_N	SPI 1 Input MOSI 1 Compliment (-) (DS_LVDS6_N)
36	SPI1_MOSI_P	SPI 1 Input MOSI 1 True (+) (DS_LVDS6_P)
39	SPI1_TR_N	SPI 1 Output Transmit Ready Compliment (-) (DS_LVDS7_N)
40	SPI1_TR_P	SPI 1 Output Transmit Ready True (+) (DS_LVDS7_P)
Shield	Ground	Ground reference is available from the connector shield

The same daughterboard used for SpaceWire (shown in Figure 2-7) can also be used for SPI interface. The interface pinout of the daughterboard is shown in Figure 2-11.

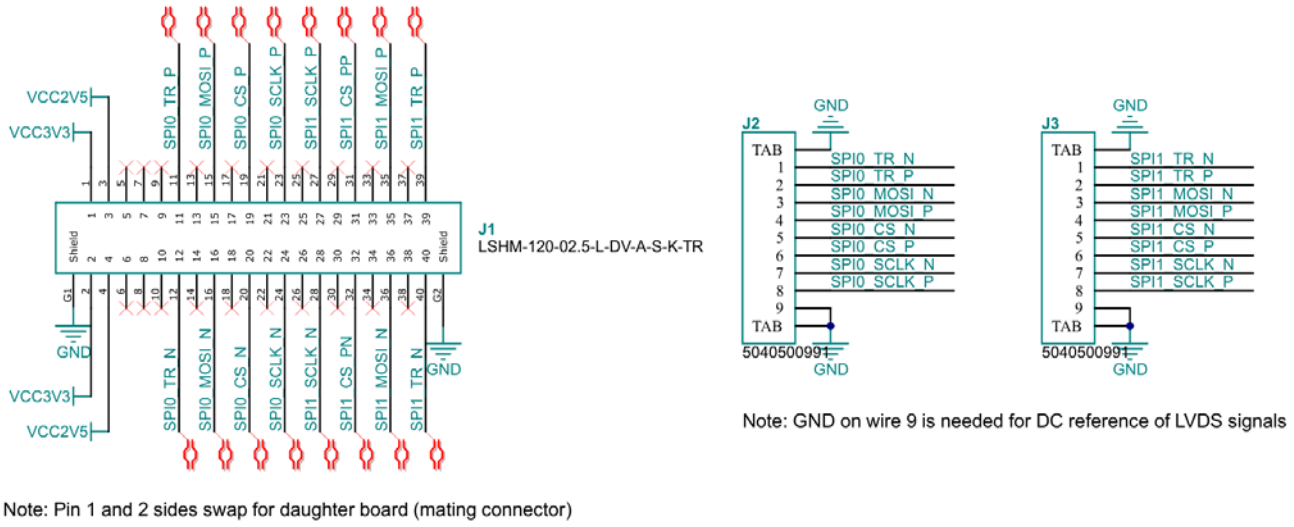


Figure 2-13: SPI interface using LSHM daughter board schematic.

2.9 RF OUTPUT

The VestaLink S25 uses a full detent SMP connector from SAMTEC, with a part number of SMP-PF-P-HE-ST-EM3.

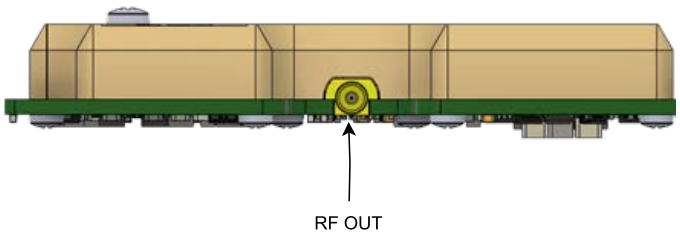


Figure 2-14: RF Connector location.

3 TT&C INTERFACE PROTOCOLS

All TT&C interfaces on the VestaLink S25 accesses the same register map, described in its user manual, regardless of the interface and/or protocol used.

3.1 I²C INTERFACE

The VestaLink S25 has a standard I²C interface for controlling the radio.

3.1.1 I²C WRITE EXAMPLE

An I²C write transaction starts with the VestaLink S25's 7-bit I²C address (0x26 by default), and the R/W bit set to 0. After that follows the register address which is 2 bytes long and then the controller must output the 4 data bytes. All registers are 4 bytes long. The register data is transferred least significant byte first. The VestaLink S25 will ACK the data and register bytes, but this only means they have been accepted.

Table 3-1 below shows an example of writing the value 8250000 (0x007DE290) to the Frequency register (address 0x0062).

Table 3-1: I²C Write Example

I²C Address & RW Bit	Addr0	Addr1	Val0	ValD1	ValD2	D3
0x26 & 0 bit	0x62	0x00	0x90	0xE2	0x7D	0x00

3.1.2 I²C READ EXAMPLE

An I²C read transaction starts with the 7-bit I²C address with the R/W bit set to 0. Then the register address is written. The I²C controller will issue a repeated start, then write the I²C address again, with the R/W bit set to 1. After that the VestaLink S25 will output the data of the register and the read address. The register data is transferred least significant byte (LSB) first. If the address was invalid, the VestaLink S25 will output all 0 values.

Table 3-2 below shows an example of reading the Temp_0 register (address 0x0032). The read register value is 0x00800079, which contains two temperature values of 32°C and 30.25°C.

Table 3-2: I²C Read Example

I²C Address & RW Bit	Addr0	Addr1
0x26 & 0 bit	0x32	0x00

I²C Address & RW Bit	Addr0	Addr1	Val0	Val1	Val2	Val3
0x26 & 1 bit	0x32	0x00	0x79	0x00	0x80	0x00

3.2 CAN INTERFACE

The VestaLink S25's CAN interface(s) can be configured to use one of three protocols, namely the CUBECOM Protocol, the Cubesat Space Protocol, or the CAN-TS Protocol.

Both interfaces must be configured to make use of the same protocol.

3.2.1 CUBECOM CAN PROTOCOL

The CUBECOM CAN Protocol uses exclusively extended identifiers. The fields inside the 29-bit CAN header are used to identify the message type, message ID (for message tracking), source address and destination address that messages will be sent to. Table 3-3 below shows the format of the CAN header identifier.

Table 3-3: CUBECOM CAN header identifier format

29-Bit CAN Header Identifier			
CAN Identifier A		**CAN Identifier B**	
5 Bits Message Type	6 Bits + 2 Bits Message ID	8 Bits Source	8 Bits Destination

Table 3-4 below shows all the possible message types for CUBECOM CAN messages.

Table 3-4: CUBECOM CAN message types.

Identifier	Message Type	Description
0x01	Write Request	Used to write a value in a register
0x02	Write Response	Response from the VestaLink S25 acknowledging a previous Write Request.
0x03	Write Error	Response from the VestaLink S25 in case of a Write failure.
0x04	Read Request	Used to read the value in a register
0x05	Read Response	Response from the VestaLink S25 with the requested register's data.
0x06	Read Error	Response from the VestaLink S25 if there is an error reading the register.

3.2.1.1 CUBECOM CAN WRITE EXAMPLE

Suppose a CAN controller with node address 0x31 needs to write to the VestaLink S25 with node address 0x26. To write data, the CAN controller will send a Write Request (0x01) message, with the address and register data in least-significant byte first (LSB) order.

Table 3-5 below shows an example of writing the value 8250000 (0x007DE290) to the Frequency register (address 0x0062).

Table 3-5: CUBECOM CAN register write request.

29 Bit CAN Header				**D0**	**D1**	**D2**	**D3**	**D4**	**D5**
Message Type	Message ID	Source	Destination	Addr0	Addr1	Val0	Val1	Val2	Val3
0x01	0x00	0x31	0x26	0x62	0x00	0x90	0xE2	0x7D	0x00

If the register is written successfully, the VestaLink S25 will respond with a Write Response (0x02) message as shown in Table 3-6 below.

Table 3-6: CUBECOM CAN write response.

29 Bit CAN Header				D0	D1
Message Type	Message ID	Source	Destination	Addr0	Addr1
0x02	0x00	0x26	0x31	0x64	0x00

If an invalid register address is sent, or if the register cannot be written for another reason, the VestaLink S25 will respond with a Write Error (0x03) message as shown in Table 3-7 below.

Table 3-7: CUBECOM CAN write error response.

29 Bit CAN Header				D0	D1
Message Type	Message ID	Source	Destination	Addr0	Addr1
0x03	0x00	0x26	0x31	0x64	0x00

3.2.1.2 CUBECOM CAN READ EXAMPLE

To read a value from the VestaLink S25, the CAN controller should send a Read request (0x04) message, with the register address in least-significant byte first (LSB) order.

Table 3-8 below shows an example of reading the Temp_0 register (address 0x0032). The read register value is 0x00800079, which contains two temperature values of 32°C and 30.25°C.

Table 3-8: CUBECOM CAN register read request.

29 Bit CAN Header				D0	D1
Message Type	Message ID	Source	Destination	Addr0	Addr1
0x04	0x00	0x26	0x31	0x32	0x00

If the register value is read successfully, the VestaLink S25 will respond with a Read Response (0x05) message containing the requested address, along with the value of the requested register as shown in Table 3-9 below.

Table 3-9: CUBECOM CAN register read response.

29 Bit CAN Header				D0	D1	D2	D3	D4	D5
Message Type	Message ID	Source	Destination	Addr0	Addr1	Val0	Val1	Val2	Val3
0x05	0x00	0x26	0x31	0x32	0x00	0x79	0x00	0x80	0x00

If an invalid register address is sent, or if the register cannot be read for another reason, the VestaLink S25 will respond with a Read Error (0x06) message as shown in Table 3-10 below.

Table 3-10: CUBECOM CAN register read error response.

29 Bit CAN Header				D0	D1
Message Type	Message ID	Source	Destination	Addr0	Addr1
0x06	0x00	0x26	0x31	0x32	0x00

3.2.2 CUBESAT SPACE PROTOCOL (CSP)

The VestaLink S25 implements a parameter system which is accessible via a CSP interface when the CAN interface is configured to use the CSP Protocol.

All parameters are exposed as 32-bit unsigned integers, with the parameter IDs corresponding to the register addresses as described in the user manual for the VestaLink S25 (USM-01-10022-A7).

More information regarding the Cubesat Space Protocol is available from the links below:

- https://github.com/libcsp/libcsp
- https://libcsp.github.io/libcsp/index.html

More information regarding the parameter system implemented is available from the link below:

- https://github.com/spaceinventor/libparam

Note(s):
- CSP protocol support is only applicable to software v2.1.0 onward.

3.2.3 CAN-TS PROTOCOL

When configured to use the CAN-TS protocol, the registers of the VestaLink S25 are accessible via standard telecommand and telemetry requests.

More information regarding the CAN-TS protocol is available from the link below:

- https://git.skylabs.si/dev/cants

The telecommand channels for the VestaLink S25 are listed in Table 3-11 below.

Table 3-11: CAN-TS telecommand channels.

Channel#	Channel Name	Description
0x00	Read Register	Used for register read requests and responses (ACK/NACK).
0x01	Write Register	Used for register write requests and responses (ACK/NACK).

The unsolicited channels for the VestaLink S25 are listed in Table 3-12 below.

Table 3-12: CAN-TS unsolicited telemetry channels.

Channel#	Channel Name	Description
0x00	Register Value	Used for sending register values, requested via the “Read Register” telecommand channel.

Note(s):
- CAN-TS protocol support is only applicable to software v2.3.1 onward.

3.2.3.1 CAN-TS KEEP-ALIVE MESSAGE EXAMPLE

The VestaLink S25 will periodically send keep-alive messages with a configurable interval (see the relevant user manual section for more information).

Table 3-13 below shows an example keep-alive message from a VestaLink S25 with the following values:

- FPGA temperature of 37°C (16-bit signed number with 0.1°C resolution).
- Total current draw of 160mA (16-bit unsigned number with 1mA resolution).
- Supply voltage of 12.3V (i.e. 12300mV - 16-bit unsigned number with 1mV resolution).

Table 3-13: CAN-TS keep-alive message (unsolicited telemetry).

29 Bit CAN Header				D0	D1	D2	D3	D4	D5
Destination [8]	Type [3]	Source [8]	Command [10]	Temp0	Tem1	Curr0	Curr1	Vol0	Vol1
0x01	0b001	0x26	0x00	0x72	0x01	0xA0	0x00	0x0C	0x30

Note(s):
- The temperature resolution for software v2.3.1 was 1°C as opposed to 0.1°C in all other supported versions.

3.2.3.2 CAN-TS CAN WRITE EXAMPLE

Suppose a CAN-TS-enabled device, with node address 0x31, needs to write to the VestaLink S25, with node address 0x26. To write data, the device will send a telecommand request on the Write Register channel (0x01), with the address and register data in least significant byte first (LSB) order.

Table 3-14 below shows an example of writing the value 8250000 (0x007DE290) to the Frequency register (address 0x0062).

Table 3-14: CAN-TS register write request (telecommand request).

29 Bit CAN Header				D0	D1	D2	D3	D4	D5
Destination [8]	Type [3]	Source [8]	Command [10]	Addr0	Addr1	Val0	Val1	Val2	Val3
0x26	0b010	0x31	0x01	0x62	0x00	0x90	0xE2	0x7D	0x00

If successful, the VestaLink S25 will respond with an appropriate telecommand ACK message[2].

If not, the VestaLink S25 will respond with an appropriate telecommand NACK message[2].

[2] Refer to the CAN-TS Protocol specifications for the format of the ACK/NACK messages.

3.2.3.3 CAN-TS CAN READ EXAMPLE

Since register addresses are 2 bytes (16 bits) long, they cannot be encoded into the command field of the CAN-TS message header (only 8 of the 10 bits available for channel). Hence, a register read occurs in two steps:

1. A telecommand containing the address of the register to be read is sent to the VestaLink S25. The VestaLink S25 will respond with an ACK if the request is valid, or a NACK if not[3].
2. If the request was valid, the VestaLink S25 will send an unsolicited telemetry message containing the address of the requested register, along with its value.

Table 3-15 below shows an example of reading the Temp_0 register (address 0x0032). The read register value is 0x00800079, which contains two temperature values of 32°C and 30.25°C.

Table 3-15: CAN-TS register read request (telecommand request).

29 Bit CAN Header				**D0**	**D1**
Destination [8]	Type [3]	Source [8]	Command [10]	Addr0	Addr1
0x26	0b010	0x31	0x00	0x32	0x00

Table 3-16 below shows the follow-up unsolicited telemetry message sent in response to a register read request, containing both the address and value of the requested register in least-significant byte first (LSB) order.

Table 3-16: CAN-TS register read response (unsolicited telemetry).

29 Bit CAN Header				**D0**	**D1**	**D2**	**D3**	**D4**	**D5**
Destination	Type	Source	Command	Addr0	Addr1	Val0	Val1	Val2	Val3
0x31	0b001	0x26	0x00	0x32	0x00	0x79	0x00	0x80	0x00

[3] Refer to the CAN-TS Protocol specifications for the format of the ACK/NACK messages.

3.3 UART INTERFACE

The UART protocol is very similar to the CUBECOM CAN protocol, with two exceptions:

1. the header is just serialized into the data stream, and
2. the source and destination addresses are reversed.

Table 3-17 below shows all the possible UART message types.

Table 3-17: UART message types.

Identifier	Message Type	Description
0x01	Write Request	Used to write a value in a register
0x02	Write Response	Response from the VestaLink S25 acknowledging a previous Write Request.
0x03	Write Error	Response from the VestaLink S25 in case of a Write failure.
0x04	Read Request	Used to read the value in a register
0x05	Read Response	Response from the VestaLink S25 with the requested data
0x06	Read Error	Response from the VestaLink S25 if there is an error reading the register.

3.3.1 UART WRITE EXAMPLE

Table 3-18 shows an example of writing the value 8250000 (0x007DE290) to the Frequency register (address 0x0064).

Table 3-18: UART register write request.

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
Message Type	Message ID	Destination	Source	Addr0	Addr1	Val0	Val1	Val2	Val3
0x01	0x00	0x26	0x31	0x64	0x00	0x90	0xE2	0x7D	0x00

If the requested register was written successfully, the VestaLink S25 will respond with a Write Response (0x02) message as shown in Table 3-19 below.

Table 3-19: UART register write response.

D0	D1	D2	D3	D4	D5
Message Type	Message ID	Destination	Source	Addr0	Addr1
0x02	0x00	0x31	0x26	0x62	0x00

If an invalid register address is sent, or if the register cannot be read for another reason, the VestaLink S25 will respond with a Write Error (0x03) message as shown in Table 3-20 below.

Table 3-20: UART register write error response.

D0	D1	D2	D3	D4	D5
Message Type	Message ID	Destination	Source	Addr0	Addr1
0x03	0x00	0x31	0x26	0x62	0x00

3.3.2 UART READ EXAMPLE

Table 3-21 below shows an example of reading the Temp_0 register (address 0x0032). The read register value is 0x00800079, which contains two temperature values of 32°C and 30.25°C.

Table 3-21: UART register read request.

D0	D1	D2	D3	D4	D5
Message Type	Message ID	Destination	Source	Addr0	Addr1
0x04	0x00	0x26	0x31	0x32	0x00

The VestaLink S25 will respond with a message containing the requested address, along with the value of the requested register as shown in Table 3-22 below.

Table 3-22: UART register read response.

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
Message Type	Message ID	Destination	Source	Addr0	Addr1	Val0	Val1	Val2	Val3
0x05	0x00	0x31	0x26	0x32	0x00	0x32	0x00	0x79	0x00

If an invalid register address is sent, or if the register cannot be read for another reason, the VestaLink S25 will respond with a Read Error (0x06) message type as shown in Table 3-23 below.

Table 3-23: UART register read error response.

D0	D1	D2	D3	D4	D5
Message Type	Message ID	Destination	Source	Addr0	Addr1
0x06	0x00	0x31	0x26	0x32	0x00

3.4 RS485/RS422 INTERFACE

The protocol for the RS485/RS422 interface is still TBD at the time of writing this document.

4 DATA INTERFACE PROTOCOLS

4.1 SPACEWIRE INTERFACE

The VestaLink S25 supports a SpaceWire interface as a data interface. Current testing has been conducted up to 200Mbps. The link is initiated by the source device at 10Mbps. The link switches over to 200Mbps after successfully negotiating the link. The maximum link rate is in-flight configurable, up to 200 Mbps.



Figure 4-1: SpaceWire link established timing simulation.

The SpaceWire link has flow control built into the protocol. Back pressure is applied to the data source if the transmitter receive FIFO is full.

4.2 CUBECOM LVDS SERIAL INTERFACE

The control of the LVDS is described in more detail in the register map section of the user manual. The overview of operation of the LVDS interface is shown in Figure 4-2. To start the link a valid clock must be detectable on the LVDS_CLK line the status signal Act_Clk will indicate if a valid clock has been detected. When the LVDS clock is active a sync word must be continuously sent to establish a bit synchronized LVDS connection. When several consecutive sync words have been captured the LVDS interface will set the Synced status signal to high. When the interface is synced, the data can be transferred from the data source to the transmitter. The FC signal will indicate that the node is ready to receive data. Any data sent while the node is not ready (FC low) and the receive buffer is full will be ignored. The node contains a receive buffer (1024 bytes) and will indicate with the flow control that the node is not ready for data if the buffer contains more than 512 bytes.

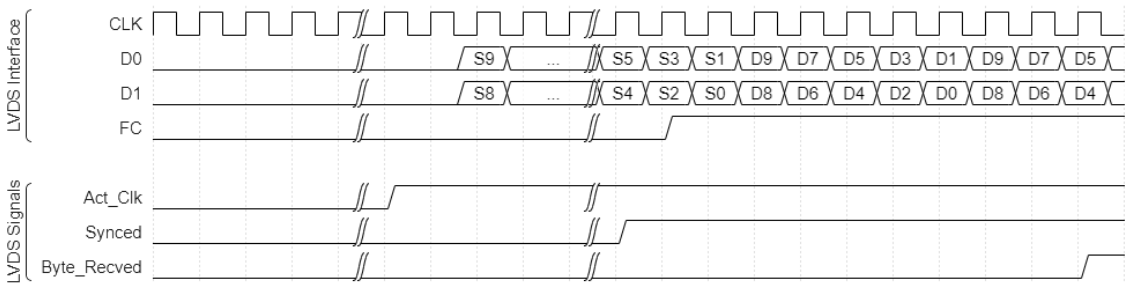


Figure 4-2: LVDS interface and status signals.

The 10-bit packet is made up of a parity bit, a control bit and a data byte. The data structure is shown in Figure 4-3. The parity bit is used to determine if a valid packet has been received. The parity bit covers 9 bits of the packet and is used to produce odd parity. Including the parity bit the packet must contain an odd number of 1's to be considered a valid packet. The control bit indicates if the packet is a control packet (C = 1) or data packet (C = 0). If an invalid packet is received the flow control will be de-asserted and the link will return to an unsynchronized state. The link must be reestablished to continue sending data.

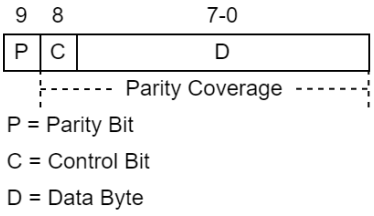


Figure 4-3: LVDS data packet structure.

The control words are provided in Table 4-1. The sync command is used to synchronize the link and to act as idle padding to keep sync while the transmitter is not ready for data. The stop command indicates the end of a "file". The transmitter has various subsections that use the EOF trigger to flush buffers or end transactions.

Table 4-1: LVDS Control Commands.

Command	Data Byte	Control Bit	Parity Bit	Control Word
Sync	0xEA	0b1	0b1	0x03EA (0b1111101010)
Stop (EOF)	0xC4	0b1	0b1	0x03C4 (0b1111000100)

The interface can be configured for a variety of different data structures. Some examples of the configurations are shown in Figure 4-4. The maximum data rate can be achieved by configuring the data link to use both data lines in DDR mode allowing for a data rate that is 4 times that of the LVDS clock rate.

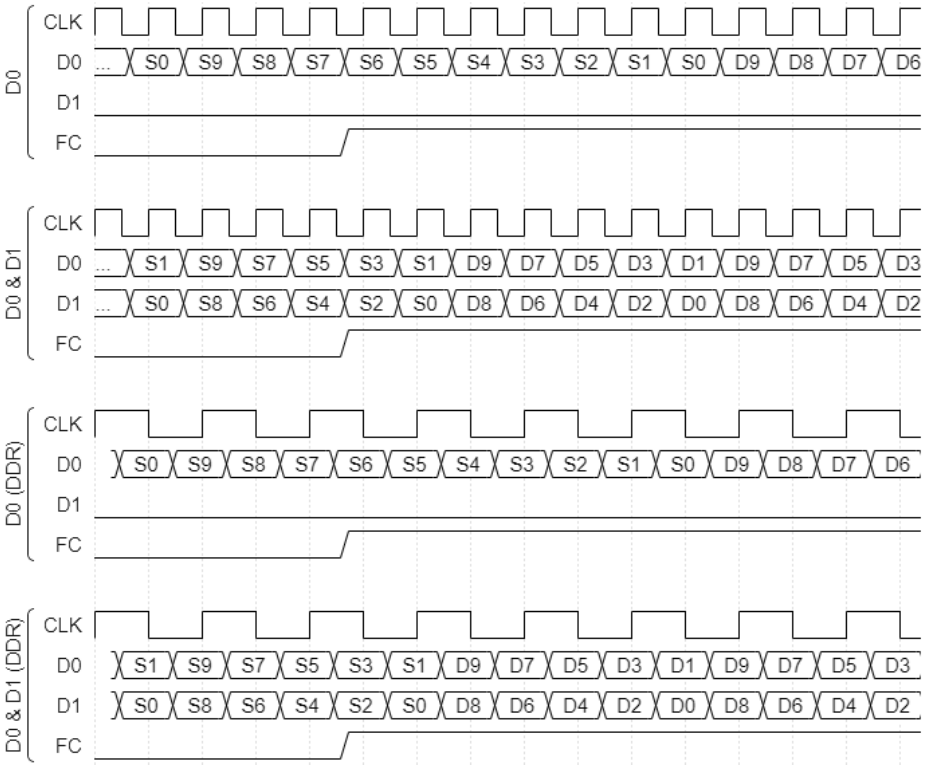


Figure 4-4: LVDS data link configuration.

4.3 SPI DATA INTERFACE

The SPI data interface supports Motorola serial protocol (Mode 1 to 4 shown in Figure 4-5). The interface consists of the standard master transmit SPI lines (SCLK, CS, MOSI). An additional flow control line (TR) is used to indicate when half the buffer is full.

The SPI interface allows for the transfer of 32-bit words of data. The data is written into a buffer (16400 bytes/4100 words). The TR line indicates that the buffer can accept another 2050 words without any data loss.

An example of the different Motorola modes of operation is shown in

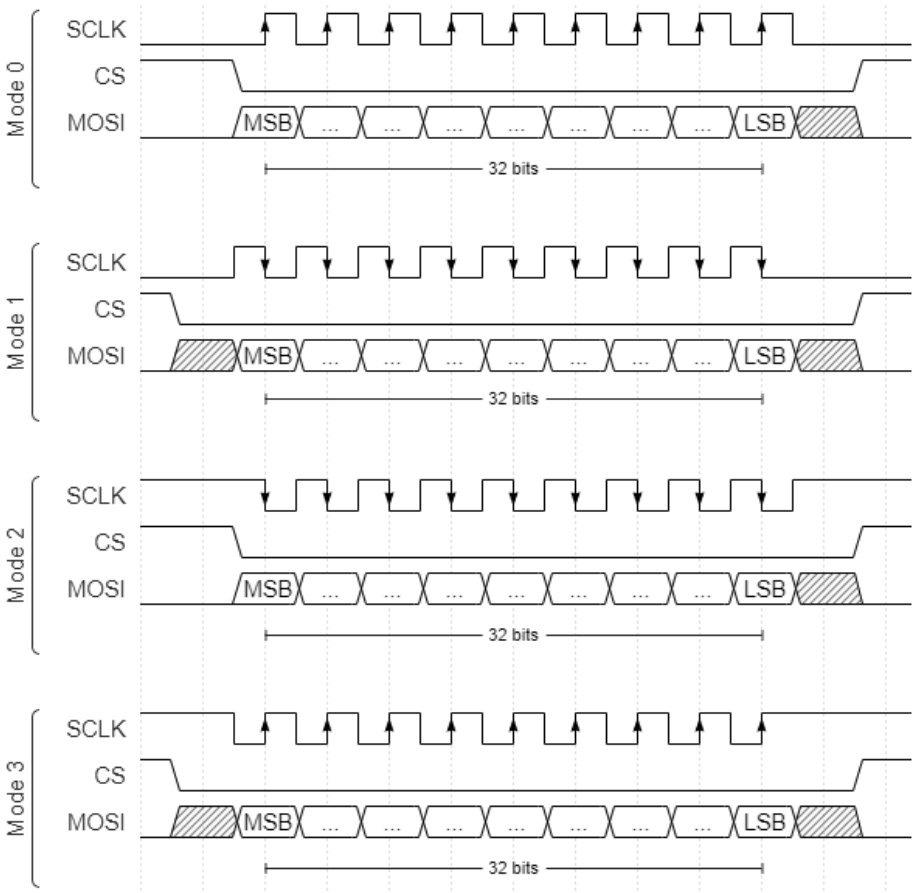


Figure 4-5: SPI Modes.

5 MECHANICAL INTERFACE

This section provides drawings detailing the mechanical interfaces of the transmitter.

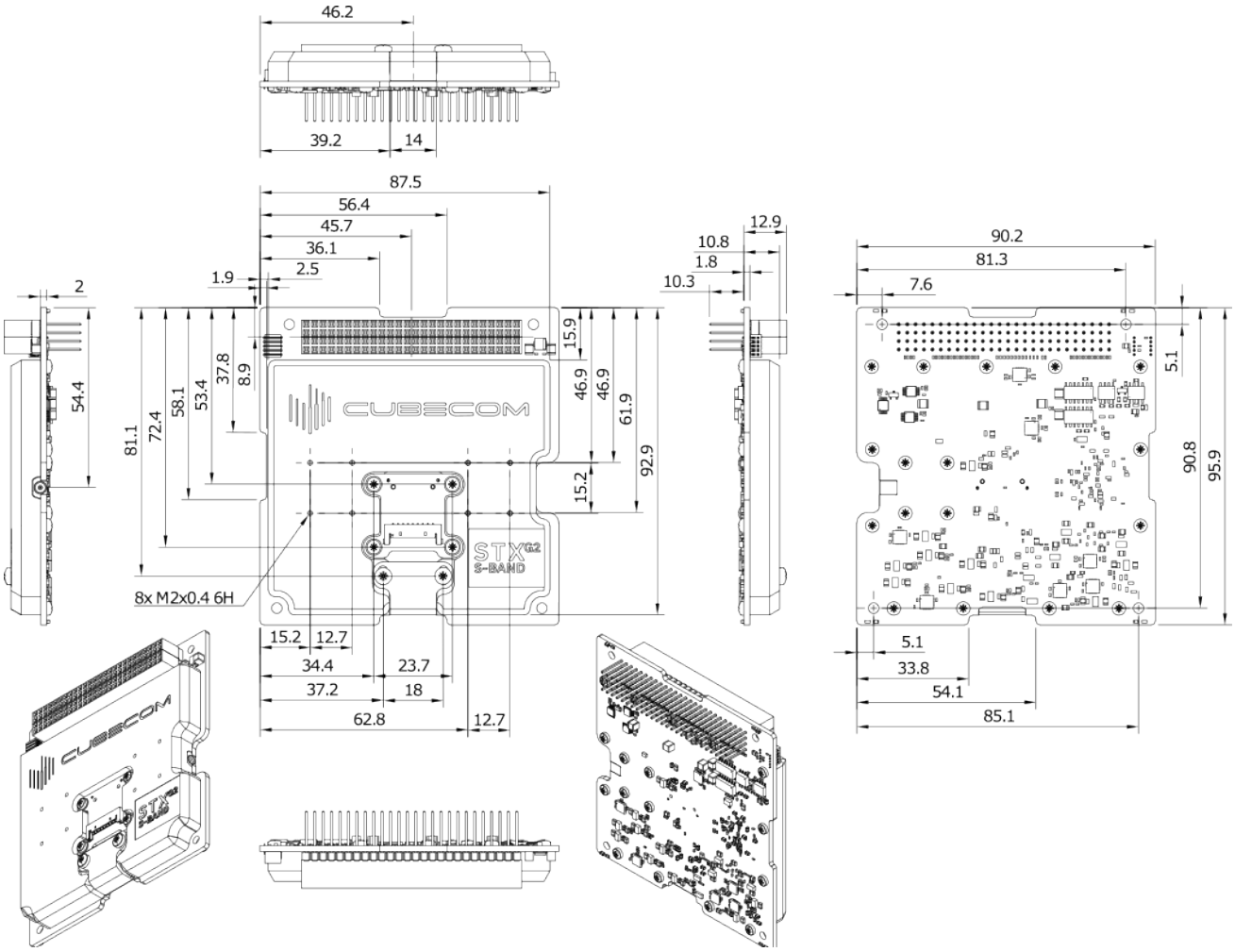


Figure 5-1: Mechanical drawing of VestaLink S25 Gen2 (PC104 connector option).

All dimensions are in millimetres, and the tolerances are as follows:

- 0.1 mm to 10 mm: +/- 0.05 mm
- 10 mm to 150 mm: +/- 0.1 mm

For any information not indicated on the diagram, please refer to the 3D data for the product.

6 THERMAL INTERFACE

The VestaLink S25 can dissipate up to 12 W of heat when transmitting 2 W of RF. The enclosure was designed to efficiently conduct the heat from the board to the heat strap locations. The heat is evenly generated across the board except for the PA which dissipates up to 5.7 W of heat. The VestaLink S25 has two options to implement thermal straps for heat dissipation. The mounting point on the right side is located directly above the PA and will have the least thermal resistance from the PA to the heat-strap.

Please not that CUBECOM do not supply the heat strap for the VestaLink S25, but it is recommended to use heat straps with a thermal interface material such as thermal pads or thermal paste for optimal performance.

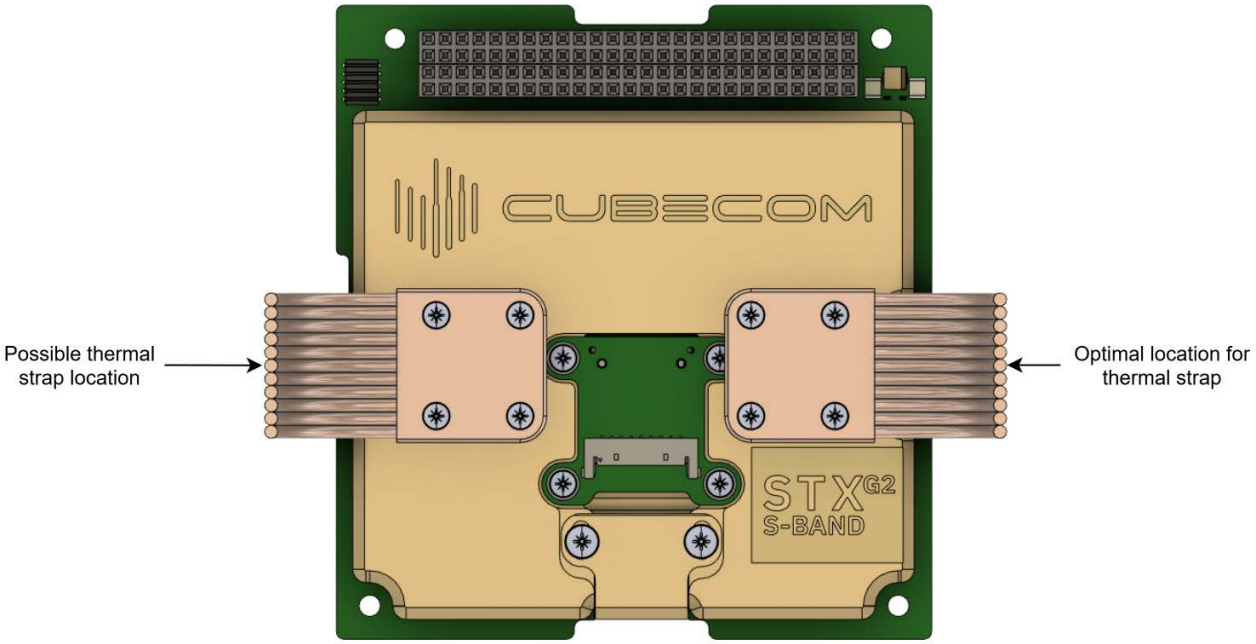


Figure 6-1: Heat-strap locations.

DOCUMENT CONTROL AND APPROVAL

Rev	Description of Change	Author	Approved By	Date
1	First draft. Mechanical drawing to be added.	S Pretorius	T Heidemann	2023/06/08
A	Updated mechanical drawings.	S Pretorius	T Heidemann	2023/06/21
A1	Added missing table references.	S Pretorius	T Heidemann	2023/11/06
A2	Updated LVDS pinout.	S Pretorius	T Heidemann	2024/01/12
A3	Updated data interfaces	S Pretorius	T Heidemann	2024/03/25
A4	Added Dual HS Interface information and diagrams	S Pretorius	T Heidemann	2024/09/04
A5	Updated Mechanical Interface drawing.	S Pretorius	T Heidemann	2024/12/02
A6	Product name updated. Updated Template, updated data interface sections, update LVDS section. Document structure updated. Added CSP & CANTS to CAN protocols section. Corrected LVDS dual node DDR entry in Table 2-13. Fixed Frequency register address from 0x0064 to 0x0062 in "CAN Interface" section. Corrected LVDS speeds	S Pretorius A v/d Westhuizen	T Heidemann M Isebeck J Leroux	2026/06/25